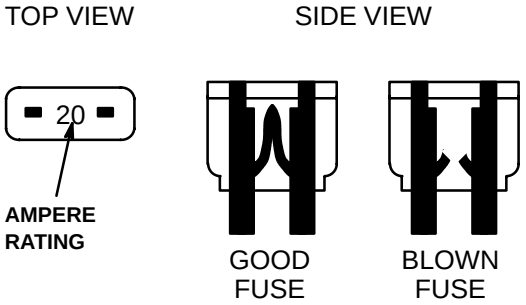
11-1 FUSE PANEL/CIRCUIT PROTECTION

CIRCUIT PROTECTION DEVICES

Electrical circuits on this vehicle may be protected by fuses, fusible links, maxi-fuses, circuit breakers, or a combination of these devices.

BLADE TYPE FUSE

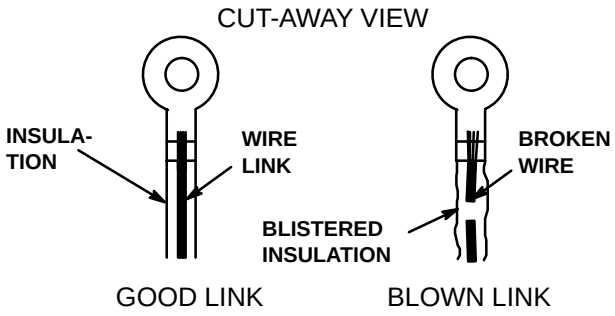


Blade type fuses have a transparent plastic housing. To check a fuse, pull it from the fuse panel and look at the fuse element through the housing. Always replace a blown fuse with a new fuse that has the same ampere rating.

The ampere rating of a blade type fuse can also be determined by following the color code shown here:

BLADE FUSE COLOR CODING	
AMPERE RATING	**HOUSING COLOR**
4	Pink
5	Tan
7.5	Brown
10	Red
15	Light Blue
20	Yellow
25	Natural
30	Light Green

FUSIBLE LINK



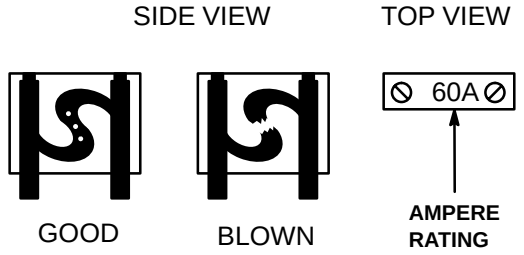
Fusible links are short lengths of wire that are smaller in diameter than the wires they are protecting. Fusible link wire is covered with a special thick, non-flammable insulation. An overload condition causes the insulation to blister. If the overload condition continues, the wire link will melt. To check a fusible link, look for blistered insulation. If the insulation is okay, pull lightly on the wire; if the fusible link stretches, the wire has melted.

When replacing fusible links, first cut the protected wire where it is connected to the fusible link. Then, tightly crimp or solder the new link to the protected wire.

Fusible links are often identified by color coding of the insulation, as shown here:

FUSIBLE LINK COLOR CODING	
WIRE LINK SIZE	**INSULATION COLOR**
20 GA	Blue
18 GA	Brown or Red
16 GA	Black or Orange
14 GA	Green
12 GA	Gray

MAXI-FUSE



Maxi-fuses have a transparent colored plastic housing. To check a maxi-fuse, look at the fuse element through the side of the housing.

To replace a maxi-fuse, pull it from the fuse box or panel. Always replace a blown maxi-fuse with a new one having the same ampere rating.

The ampere rating of a maxi-fuse can also be determined by following the color code shown here:

MAXI-FUSE COLOR CODING	
AMPERE RATING	**HOUSING COLOR**
30	Light Green
40	Amber
50	Red
60	Blue

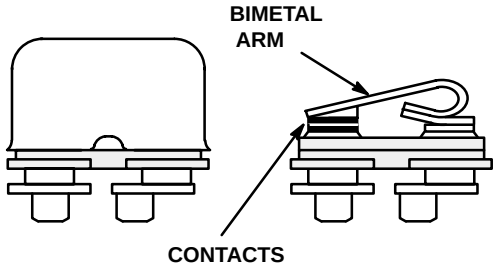
CIRCUIT BREAKER

Some circuits are protected by circuit breakers (abbreviated “c. b.” in fuse chart).
They can be Fuse Panel mounted or in-line. Like fuses, they are rated in amperes.

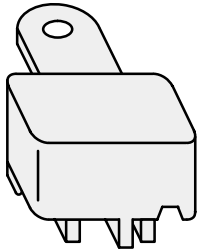
Each circuit breaker conducts current through an arm made of two types of metal bonded together (bimetal arm). If the arm starts to carry too much current, it heats up. As one metal expands faster than the other, the arm bends, the contacts open, and current flow is broken. A circuit breaker can be the cycling or non-cycling type.

FUSE PANEL MOUNTED CYCLING TYPE

SIDE VIEW (EXTERNAL) SIDE VIEW (INTERNAL)



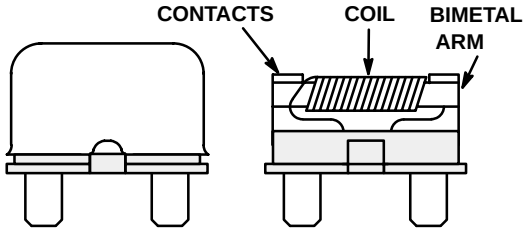
IN-LINE MOUNTED CYCLING TYPE



In the cycling type, the bimetal arm cools and straightens out. This cycle repeats as long as the overcurrent exists and power is applied.

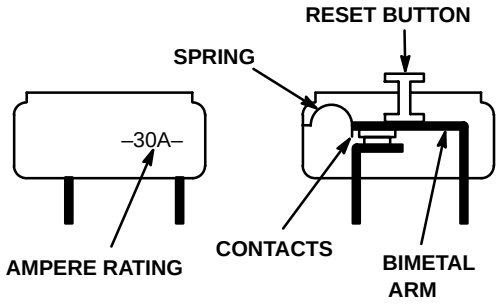
FUSE PANEL MOUNTED NON-CYCLING TYPE

SIDE VIEW (EXTERNAL) SIDE VIEW (INTERNAL)



FUSE PANEL MOUNTED MANUAL RESET TYPE

SIDE VIEW (EXTERNAL) SIDE VIEW (INTERNAL)

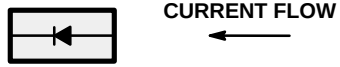


Two types of non-cycling circuit breakers are used; one is reset by removing power from the circuit, and the other is reset by depressing a reset button.

In the first type, there is a coil wrapped around the bimetal arm. When an overcurrent exists and the contacts open, a small current passes through the coil. This current through the coil is not enough to operate a load, but it does heat up both the coil and the bimetal arm. This keeps the arm in the open position until power is removed.

In the second type, a spring pushes the bimetal arm down and holds the contacts together. When an overcurrent condition exists and the bimetal arm heats up, the bimetal arm bends enough to overcome the spring and the contacts snap open. The contacts stay open until the reset button is pushed and the contacts snap together again.

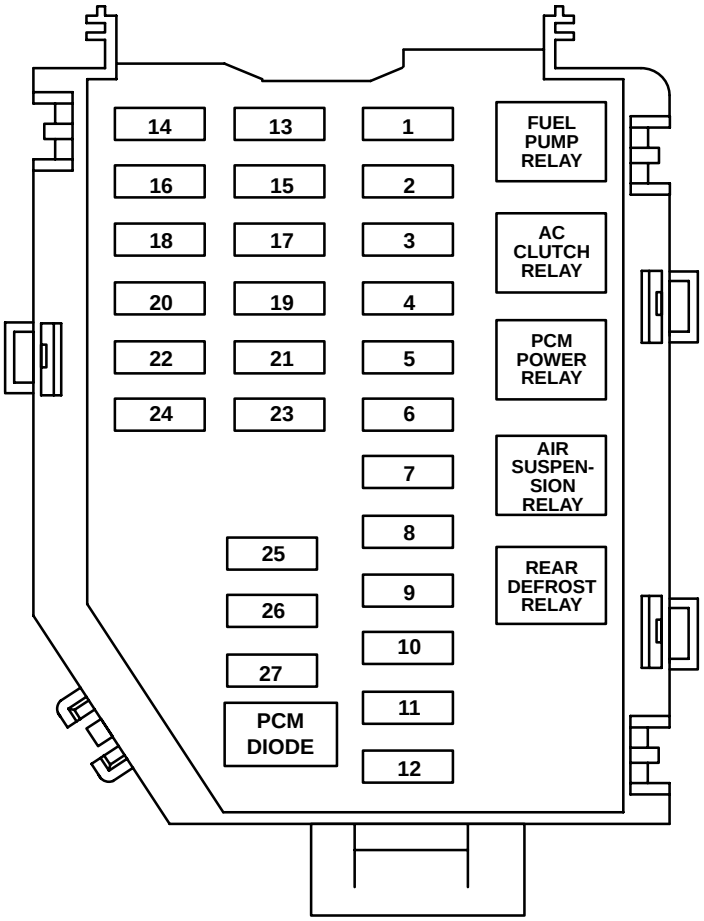
DIODE



Diodes are electrical devices that permit current to flow in one direction only. The current flows in the direction indicated by the arrow.

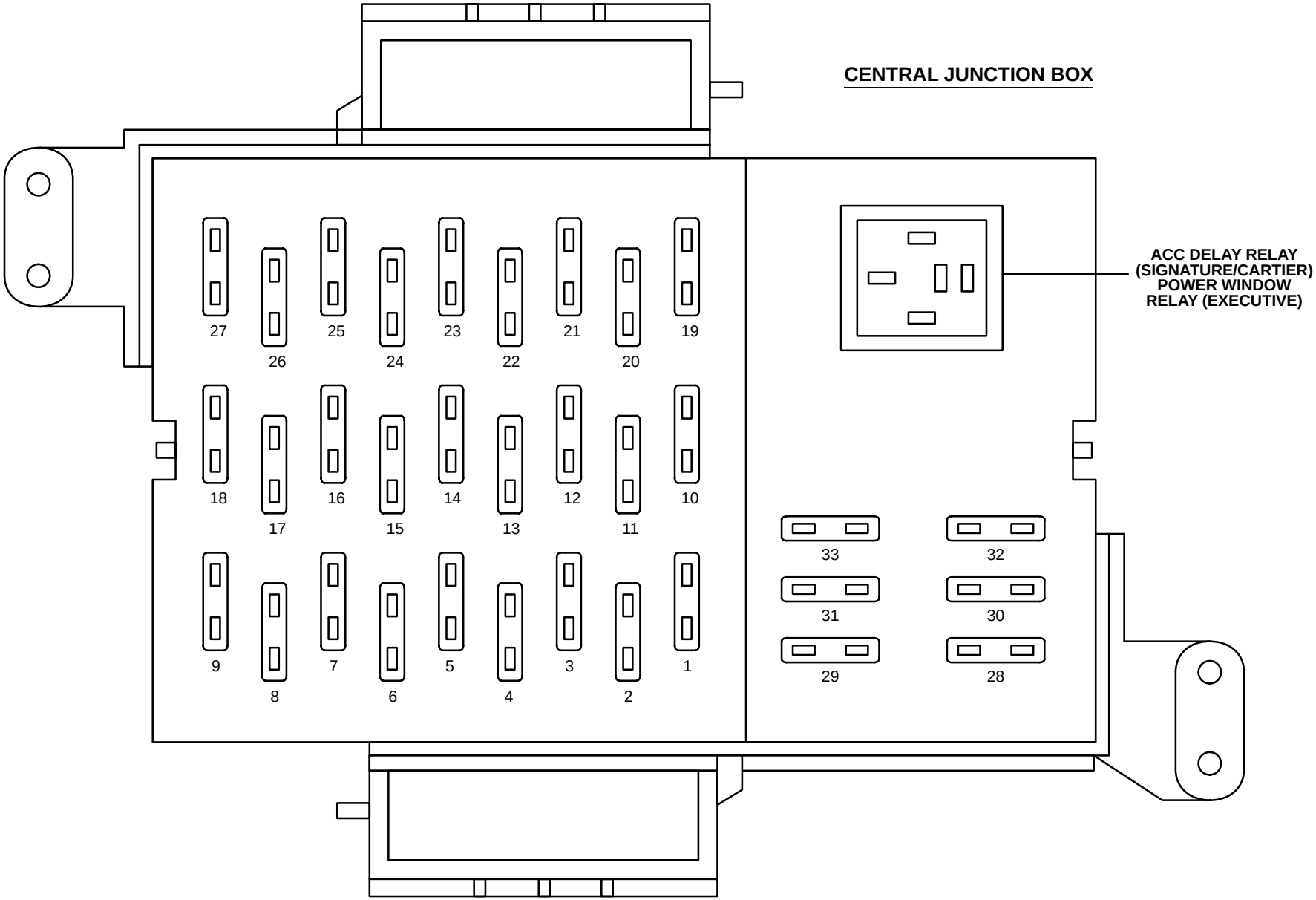
11-3 FUSE PANEL/CIRCUIT PROTECTION

2001 TOWN CAR



BATTERY JUNCTION BOX

MAXI FUSE	Amps	Circuits Protected	Page
1	50	Ignition Switch, Starter Relay, Generator	13-1
2	40	Ignition Switch	13-2
3	50	Cooling Fan High Relay	13-2
4	30	PCM Power Relay	13-2
5	40	Central Junction Box, Fuses 11, 19, 21, 23, 25, 27, 32	13-2
6	–	Not Used	–
7	40	Central Junction Box, Fuses 1, 3, 5, 7, 9, 31	13-3
8	30	Central Junction Box, Fuse 30, Power Lumbar Seats, Power Seats, Adjustable Pedal	13-3
9	40	Anti-Lock Brake System (ABS) Module	13-3
10	40	Rear Defrost Relay	13-3
11	40	Accessory Delay Relay (Signature) , Power Window Relay (Executive), Central Junction Box, Fuse 29	13-4
12	30	Compressor Relay	13-4
25	30	Passenger Power Seat (Long Wheel Base)	13-5
26	30 c.b.	Cooling Fan Low Relay	13-5
27	20	Anti-Lock Brake System (ABS) Module	13-5
MINI FUSE	**Amps**	**Circuits Protected**	**Page**
13	30	Rear Heated Seats (Long Wheel Base)	13-19
14	20	Rear Power Point (Long Wheel Base)	13-19
15	20	Rear Power Point (Long Wheel Base)	13-19
16	30	Heated Seats	13-4
17	10	Air Suspension	13-5
18	15	Horn Relay	13-5
19	30	Subwoofer Amplifier, Central Junction Box, Fuse 33	13-5
20	15	Fuel Injectors, PCM, IAC Valve, Mass Air Flow Sensor	13-6
21	15	Heated Oxygen Sensors, Trans Solenoids, EVAP Canister Vent Solenoid, EGR Vacuum Regulator (EVR), EVAP Canister Purge Solenoid, 4R70W Transmission	13-7
22	20	Fuel Pump Relay, Fuel Pump Prime Connector	13-4
23	15	Generator	13-4
24	20	Auxiliary Power Outlet	13-5


CENTRAL JUNCTION BOX
ACC DELAY RELAY (SIGNATURE/CARTIER) POWER WINDOW RELAY (EXECUTIVE)
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11-5 FUSE PANEL/CIRCUIT PROTECTION

REVISED

FUSE	Amps	Circuits Protected	Page
1	10	Lighting Control Module (LCM)	13-10
2	30	EATC Blower Motor	13-10
3	10	Lighting Control Module (LCM)	13-10
4	7.5	Instrument Cluster	13-10
5	7.5	Lighting Control Module (LCM)	13-10
6	15	EATC, Heated Seats, A/C Clutch Relay, A/C Clutch Cycling Pressure Switch	13-11
7	15	Lighting Control Module (LCM), Day/Night Sensor/Amplifier	13-10
8	10	Speed Control, Air Suspension, Steering Wheel Rotation Sensor, Shift Lock Actuator	13-11
9	20	Lighting Control Module (LCM), Multifunction Switch	13-10
10	10	Restraint Control Module (RCM), Air Bags	13-12
11	–	Not Used	–
12	15	Instrument Cluster, Anti-Theft, Ignition Coils, Radio Noise Capacitor	13-13
13	10	Anti-Lock Brake System (ABS) Module	13-12
14	7.5	Transmission Control Switch (TCS), Lighting Control Module (LCM)	13-12
15	20	Multifunction Switch	13-12
16	30	Wiper Control Module (WCM), Windshield Wiper Motor	13-14
17	10	Digital Transmission Range (DTR) Sensor	13-12
18	7.5	Lighting Control Module (LCM), Front Radio Control Unit, Cellular Telephone Transceiver, Electronic Day/ Night Mirror, Compass Module, Rear Audio/Climate Control Module	13-14
19	10	EATC, Clock, Instrument Cluster, PCM	13-15
20	7.5	Lighting Control Module (LCM), ABS, Shift Lock Actuator	13-16
21	20	Multifunction Switch	13-15
22	20	Multifunction Switch, High Mounted Stoplamps	13-16
23	20	Data Link Connector (DLC), I/P Cigar Lighter	13-15
24	5	Front Radio Control Unit	13-16
25	15	Lighting Control Module (LCM)	13-15
26	5	Digital Transmission Range (DTR) Sensor	13-16
27	20	Fuel Filler Door Release Switch	13-15
28	10	Heated Mirror Heaters	13-17
29	20	Driver Door Module	13-17
30	7.5	Driver Seat Module, Luggage Compartment Lid Release Switch, Door Lock Switches, LF Seat Control Switch Driver Door Module, Exterior Rear View Mirror Switch, Adjustable Pedal Switch, Keypad Switch Assembly	13-17
31	7.5	Main Light Switch, Lighting Control Module (LCM)	13-18
32	25	Brake Pedal Position (BPP) Switch, Deactivator Switch	13-9
33	15	Front Radio Control Unit, Digital Compact Disc (CD) Changer, Cellular Telephone Transceiver	13-18